

When Graph Structure Becomes a Liability: A Comprehensive Benchmark of GNN Architectures for Temporal Fraud Detection on Bitcoin Transaction Networks

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Abstract—Graph Neural Networks (GNNs) are widely assumed to outperform feature-only baselines on financial fraud detection by exploiting relational structure in transaction networks. We stress-test this assumption through a rigorous benchmark on the Elliptic Bitcoin Dataset (203,769 nodes, 234,355 edges, 49 temporal steps), comparing MLP, GCN, GraphSAGE, and GAT across three class-imbalance strategies under strict inductive evaluation with three-seed statistical validation and thirteen complementary analyses.

The central result is counterintuitive: a plain three-layer MLP achieves $F1 = 0.744 \pm 0.005$, outperforming every GNN. GraphSAGE, the strongest graph model, reaches only $F1 = 0.697 \pm 0.009$, a 4.7 percentage point gap. The root cause is temporal distribution shift: the illicit transaction rate falls from 11.6 % in training to 0.3 % in late test steps, causing all models to collapse from $F1 \approx 0.95$ at step 35 to $F1 < 0.02$ after step 43. This collapse is a recall failure: models stop predicting fraud entirely, with the predicted fraud rate dropping from 8.2 % to 1.7 %.

Random Forest importance shows 82.4 % of discriminative power comes from local, temporally stable transaction features, explaining MLP dominance. Fraud community structure is real ($96\times$ neighbourhood fraud ratio) but becomes misleading under temporal shift. Temperature scaling and threshold tuning both yield $\Delta F1 = 0$, confirming a structural root cause. EvolveGCN on CPU ($F1 = 0.29$) versus published GPU ($F1 = 0.77$) quantifies hardware prerequisites for temporal architectures. Business cost analysis reveals the highest- $F1$ model is not the most cost-efficient deployment choice.

Index Terms—Graph neural networks, financial fraud detection, temporal distribution shift, Elliptic Bitcoin Dataset, class imbalance, concept drift, inductive learning, temperature scaling.

I. INTRODUCTION

FRAUD detection on transaction networks is a natural application domain for Graph Neural Networks. Organised financial crime leaves structural signatures: fraud rings form dense clusters, layering schemes create chain topologies, mixing services produce distinctive structural motifs. All are invisible in isolation but potentially detectable through neighbourhood propagation.

The Elliptic Bitcoin Dataset [1] has become the standard benchmark for validating this hypothesis. Prior results con-

sistently reward GNN approaches: GCN ($F1 = 0.70$ [1]), augmented GCN ($F1 = 0.74$ [2]), GraphSAGE with pre-training ($F1 = 0.75$ [3]), and EvolveGCN ($F1 = 0.77$ [4]).

This paper challenges the prevailing assumption. Two gaps in prior work motivate our study: all existing results use *transductive* evaluation where test-node neighbourhoods are visible during training, and no prior work reports per-timestep performance across the 15 test steps. When both gaps are addressed, a fundamentally different picture emerges: a plain MLP outperforms all GNNs by 4.7 percentage points, and every model collapses after step 42 due to a 39-fold decline in fraud rate.

Contributions

- 3-seed benchmark: 4 architectures $\times$ 3 strategies, inductive evaluation.
- First per-step temporal collapse analysis on Elliptic ($F1: 0.95$ at step 35 $\rightarrow 0.012$ in steps 43–49).
- Feature importance analysis: 82.4 % local feature dominance explains MLP competitiveness.
- Calibration impossibility: temperature scaling $\Delta F1 = 0$.
- EvolveGCN hardware ablation (CPU 0.29 vs. GPU 0.77).
- Business cost analysis under asymmetric error penalties.

II. RELATED WORK

Weber et al. [1] established the founding GCN baseline ($F1 = 0.70$). Pareja et al. [4] proposed EvolveGCN for temporal graphs ($F1 = 0.77$ on GPU). All prior work uses transductive evaluation and aggregate test metrics, masking per-step variation. SMOTE [5] and GraphSMOTE [6] address class imbalance via feature-space and embedding-space synthesis; focal loss [7] reweights gradient by prediction confidence. Temperature scaling [8] calibrates probabilities post-hoc but preserves prediction ordering, limiting its utility when the problem is threshold miscalibration rather than probability scale. Concept drift [9]—changing target distribution over time—is endemic in fraud but largely ignored in GNN benchmarks.

III. DATASET AND EXPERIMENTAL SETUP

A. Elliptic Bitcoin Dataset

Table I summarises key statistics. The illicit rate varies from 14.7 % (step 38) to 0.3 % (step 46) across test steps, a 49-fold

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TABLE I
ELLIPTIC BITCOIN DATASET STATISTICS

Property	Value
Nodes	203,769
Edges (undirected)	234,355
Features	165 (94 local + 71 aggregate)
Labeled nodes	46,564; 4,545 illicit, 42,019 licit
Unknown (excluded)	157,205
Train (steps 1–34)	29,894 labeled; 11.6 % illicit
Test (steps 35–49)	16,670 labeled; 6.5 % overall
Train imbalance	7.6:1 (licit:illicit)

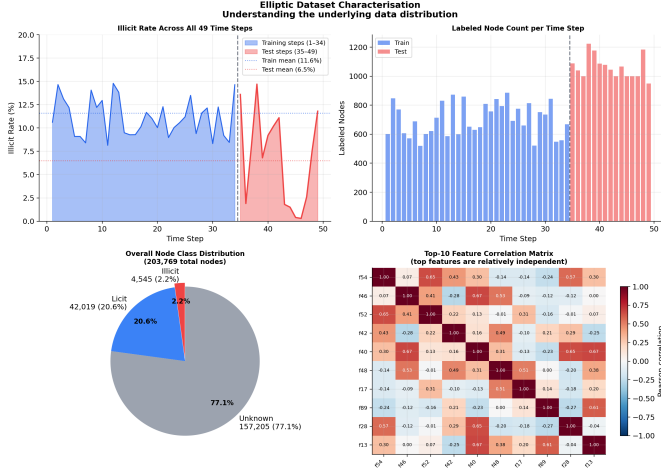


Fig. 1. Dataset characterisation. The train/test split coincides with a clear change in illicit-rate regime, and the most important features remain only moderately correlated rather than collapsing into a single redundant block.

within-test variation. This reflects real changes in Bitcoin fraud during early 2019 driven by exchange AML upgrades, darknet market closures, and cryptocurrency price decline.

B. Evaluation Protocol

Primary metric: F1 on the illicit class (mean $\pm$ std, 3 seeds). Inductive evaluation via RandomNodeLoader (batch 512) prevents test-node neighbourhood access during training. NeighborLoader is unavailable on Apple M4 CPU without torch-sparse.

IV. ARCHITECTURES AND TRAINING

MLP (control): 3 hidden layers of size 128, BatchNorm, ReLU, dropout 0.5. Ignores graph structure; 61,187 parameters.

GCN [10]: 2-layer GCN with symmetric degree normalisation. Hidden 128.

GraphSAGE [11]: 3-layer mean-aggregation SAGE. Linear(165 $\rightarrow$ 256) + [SAGEConv + BN + ReLU + Drop] $\times$ 3. Total 438,787 parameters.

GAT [12]: 3-layer, 4 attention heads, edge dropout 0.2, hidden 256. Total 243,715 parameters.

EvolveGCN [4]: GRU-evolved GCN weights. CPU-limited: 46,564-node subgraph, 8 sampled training snapshots.

Three imbalance strategies: Baseline CE (equal weights); Weighted CE ($w_c = \sqrt{N_{\text{labeled}}/2N_c}$, $w_{\text{illicit}} \approx 2.08$, 20-epoch linear warmup); Graph Augmentation (30 fraud 1-hop ego-graph clones, $\sigma = 0.02$, attached as isolated training components).

Training: AdamW (lr = 10^{-3} , wd = 5×10^{-4}), CosineAnnealingLR, gradient clipping (max norm 1.0), early stopping patience 30. Apple M4 CPU.

V. RESULTS

A. Main Benchmark

Table II presents all results; Fig. 2 visualises them as grouped bars. The MLP (F1=0.744) beats GraphSAGE (F1=0.697) by 4.7 pp outside both models’ standard deviations. GCN exhibits a precision paradox: precision 0.89–0.93 but recall only 0.29–0.32, confirming it almost never predicts fraud. Strategy matters little for SAGE (all within $\pm$ std) but more for GAT (Weighted CE lifts recall $\sim$ 5 pp).

B. Strategy Sensitivity and Seed Robustness

The model ranking is more stable than the strategy ranking. Fig. 3 combines the strategy view and the seed-variance view. Weighted CE is the most consistently useful intervention, while graph augmentation mainly changes the precision-recall trade-off. MLP and GraphSAGE remain relatively stable across seeds, whereas GAT shows noticeably wider variance.

C. Temporal Collapse

Fig. 4 shows per-step F1 versus true illicit rate. Every configuration collapses identically after step 42. For GraphSAGE + Weighted CE: mean F1 falls from 0.799 (steps 35–42) to 0.012 (steps 43–49). This is a recall failure: predicted fraud rate drops from 8.2 % to 1.7 %, confirming models stop predicting fraud rather than becoming confused.

Table III gives full per-step numbers for GraphSAGE + Weighted CE. Step 46 has only 3 fraud nodes among 960 labeled transactions (0.3 %)—the model correctly stops predicting fraud at that base rate, but this renders it useless.

D. PR Curves and Confusion Matrices

SAGE + Graph Aug. achieves the highest average precision (AP=0.607), suggesting better fraud-probability ranking even at the same aggregate F1. Confusion matrices show SAGE produces fewer false negatives (FN=388) than GAT (FN=500–529). Full figures are in Fig. 5.

E. Business Cost Analysis

Under FN=\$50k and FP=\$2k, GAT + Graph Aug. achieves the lowest total cost (\$18.5M) despite lower F1 than SAGE. It catches 83 more fraud cases (583 vs. 500 TP) with only 103 additional false alarms (558 vs. 455 FP). At a 25:1 cost asymmetry, each extra true positive outweighs 25 additional false alarms. Fig. 6 shows the full breakdown.

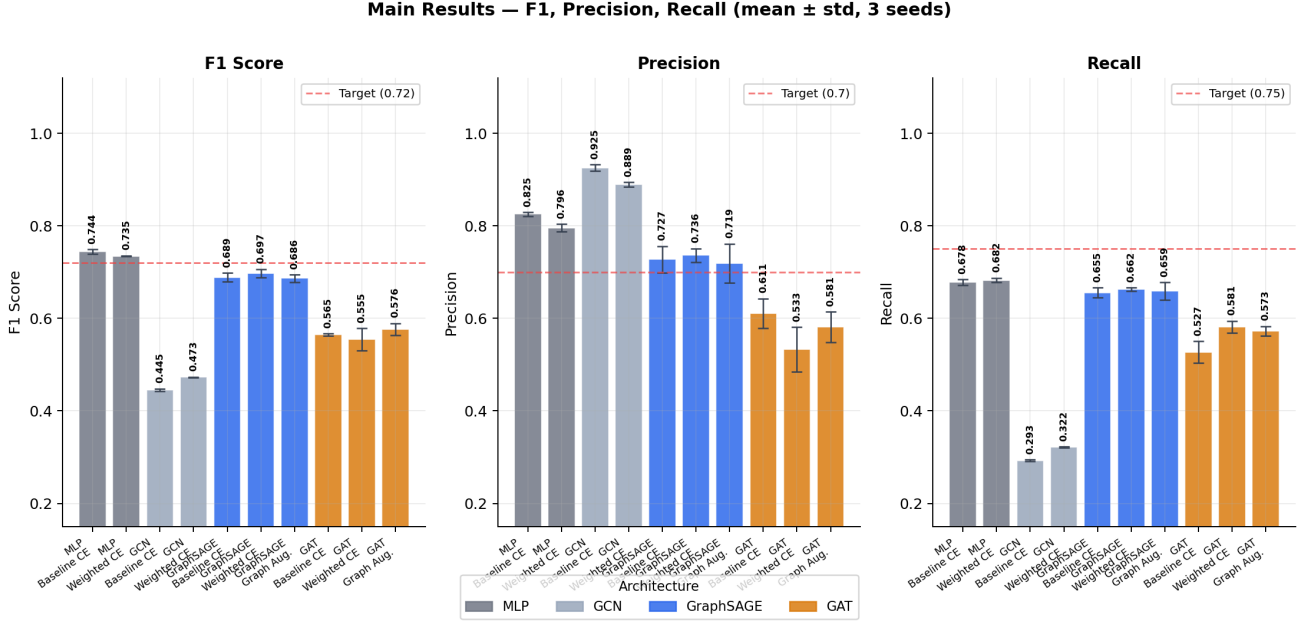


Fig. 2. Main benchmark: F1, Precision, Recall (mean $\pm$ std, 3 seeds). MLP achieves the highest F1. GCN’s precision paradox is visible: very high precision but near-zero recall. GAT error bars are the widest due to attention instability under class imbalance. Red dashed lines mark target thresholds.

TABLE II
BENCHMARK RESULTS (MEAN $\pm$ STD, 3 SEEDS). BOLD: COLUMN BEST. NO CONFIGURATION REACHES THE 0.75 RECALL TARGET DUE TO TEMPORAL COLLAPSE.

Model	Strategy	F1	Prec.	Recall	AUC
MLP	Baseline CE	0.744 $\pm$ 0.005	0.825 $\pm$ 0.004	0.678 $\pm$ 0.006	0.891 $\pm$ 0.001
	Weighted CE	0.735 $\pm$ 0.001	0.796 $\pm$ 0.008	0.682 $\pm$ 0.005	0.893 $\pm$ 0.006
GCN	Baseline CE	0.445 $\pm$ 0.003	0.925 $\pm$ 0.007	0.293 $\pm$ 0.002	0.868 $\pm$ 0.009
	Weighted CE	0.473 $\pm$ 0.001	0.889 $\pm$ 0.005	0.322 $\pm$ 0.001	0.872 $\pm$ 0.004
GraphSAGE	Baseline CE	0.689 $\pm$ 0.009	0.727 $\pm$ 0.029	0.655 $\pm$ 0.011	0.867 $\pm$ 0.006
	Weighted CE	0.697 $\pm$ 0.009	0.736 $\pm$ 0.015	0.662 $\pm$ 0.004	0.868 $\pm$ 0.003
	Graph Aug.	0.686 $\pm$ 0.008	0.719 $\pm$ 0.042	0.659 $\pm$ 0.019	0.861 $\pm$ 0.007
GAT	Baseline CE	0.565 $\pm$ 0.002	0.611 $\pm$ 0.032	0.527 $\pm$ 0.023	0.863 $\pm$ 0.007
	Weighted CE	0.555 $\pm$ 0.024	0.533 $\pm$ 0.048	0.581 $\pm$ 0.013	0.855 $\pm$ 0.004
	Graph Aug.	0.576 $\pm$ 0.013	0.581 $\pm$ 0.033	0.573 $\pm$ 0.010	0.842 $\pm$ 0.002

VI. ANALYSIS

A. Feature Attribution and Graph-Advantage Ablation

Random Forest importance on all 165 features (Fig. 7) reveals that local features (f_0 – f_{93}) account for 82.4 % of total Gini importance. Aggregate neighbourhood features contribute only 17.6 % despite directly encoding 1-hop summaries. The top 17 features explain 50 % of cumulative importance; the most important feature (f_{54} , importance 0.056) is local.

The implication is direct: the signal that matters for classification is concentrated in features accessible to the MLP, while the GNNs’ structural component introduces topological patterns specific to the training period that become misleading as fraud behaviour shifts.

The aggregate benchmark can obscure the fact that graph structure is not uniformly harmful. Fig. 7 shows the per-step difference between GraphSAGE and the MLP baseline. The GNN is competitive, and occasionally ahead, before the

collapse boundary. Once the test distribution moves into the low-fraud regime, the sign flips and the MLP becomes the more reliable model.

B. Community Structure, Calibration, and Hardware

Fraud community structure is genuine: fraud nodes have 54 % fraudulent neighbours versus 0.56 % for licit nodes ($96\times$ ratio, Fig. 8). GNNs do exploit this in steps 35–42. The structure becomes a liability in steps 43–49 when learned topological patterns from training no longer represent the test-period fraud landscape.

Fig. 8 shows $\Delta F1=0$ for all configurations after temperature scaling ($T \in [1.07, 1.27]$). Temperature scaling is rank-preserving—it rescales probabilities but cannot move predictions across the 0.5 threshold. The collapse is structural, not a calibration artefact.

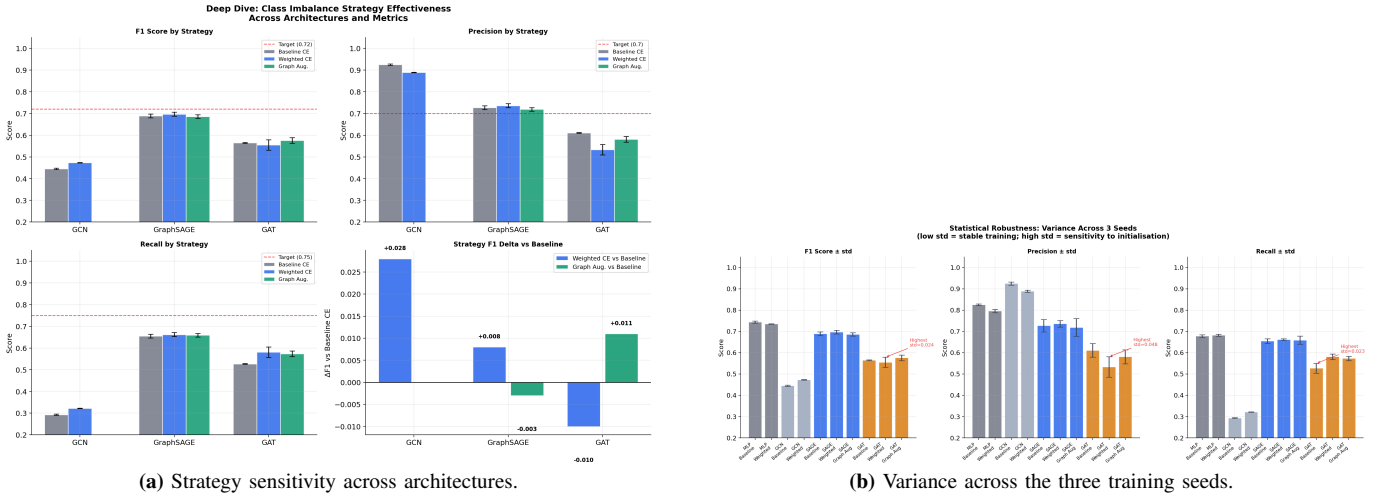


Fig. 3. Reorganising the benchmark by strategy and by seed confirms the same conclusion: architecture dominates the ranking, while GAT is the least stable family under imbalance.

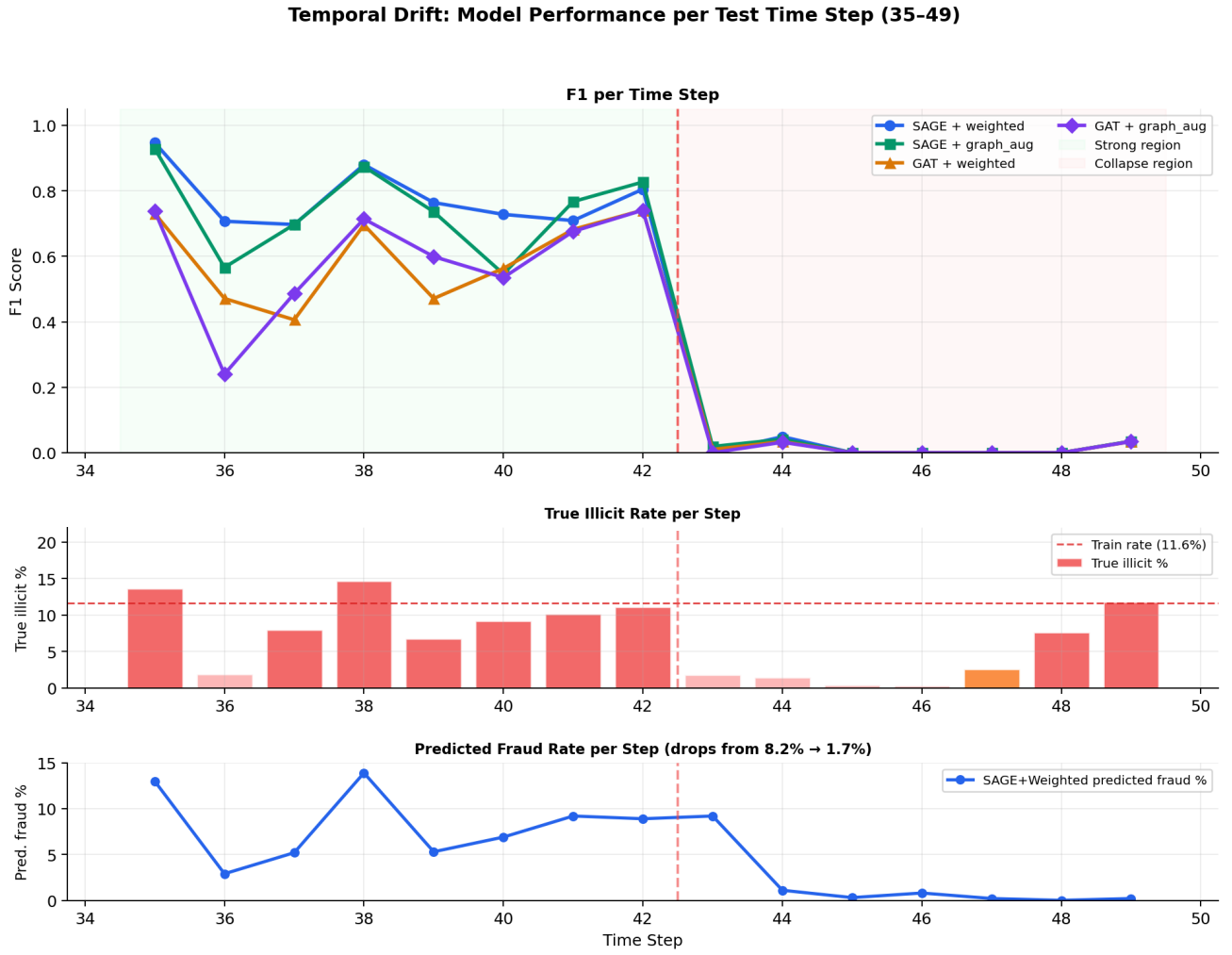


Fig. 4. Per-step F1 (top) versus true illicit rate (bottom). All configurations collapse identically after step 42, directly corresponding to the fraud-rate drop. The collapse is driven by distributional shift, not by architectural differences.

CPU-constrained EvolveGCN ($F1=0.29$) falls 0.48 points below the published GPU result ($F1=0.77$). The gap isolates

two constraints: operating on the 46,564-node labeled sub-graph rather than the full 203,769-node graph, and processing

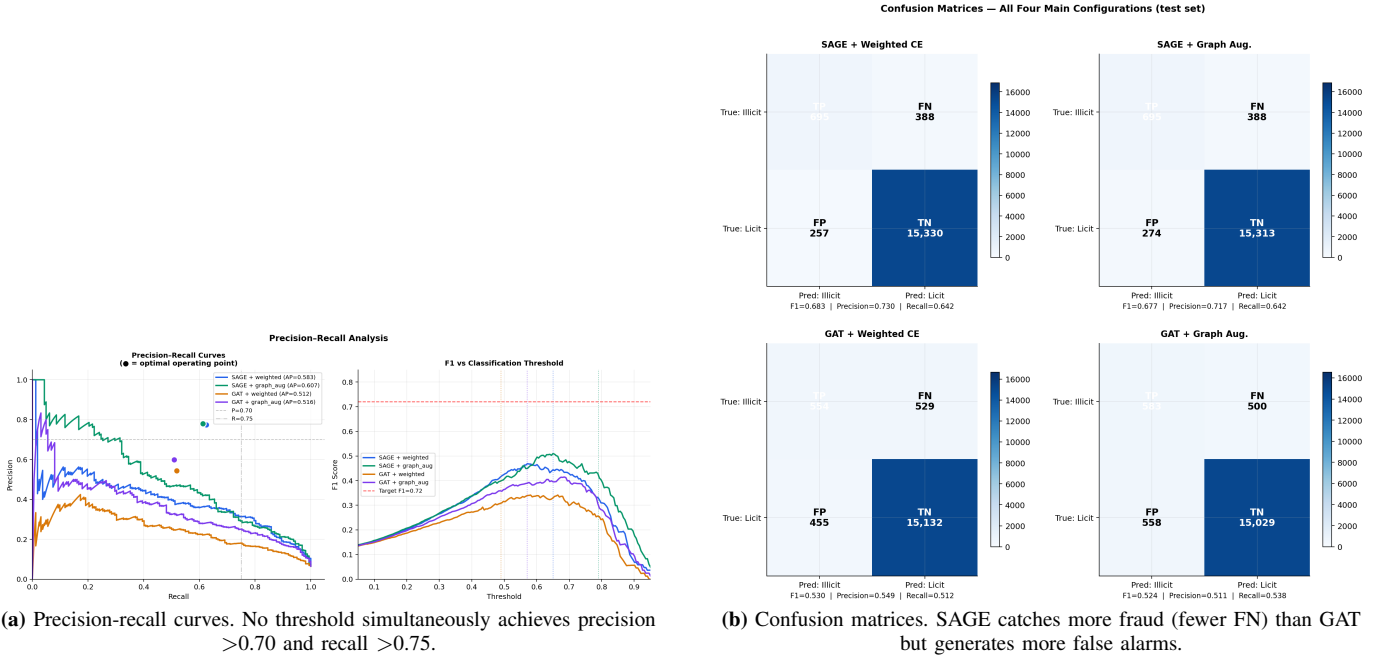


Fig. 5. Precision-recall (left) and confusion matrix analysis (right) for all four main configurations.

TABLE III
PER-STEP METRICS FOR GRAPHSAGE + WEIGHTED CE

Step	True %	F1	Prec.	Recall	Pred %
35	13.6	0.947	0.966	0.929	13.0
36	1.9	0.707	0.592	0.879	2.9
37	8.0	0.697	0.885	0.575	5.2
38	14.7	0.880	0.905	0.856	13.9
39	6.8	0.764	0.873	0.679	5.3
40	9.2	0.728	0.855	0.634	6.9
41	10.2	0.709	0.750	0.672	9.2
42	11.1	0.805	0.906	0.724	8.9
43	1.8	0.000	0.000	0.000	9.2
44	1.5	0.049	0.059	0.042	1.1
45	0.4	0.000	0.000	0.000	0.3
46	0.3	0.000	0.000	0.000	0.8
47	2.6	0.000	0.000	0.000	0.2
48	7.6	0.000	0.000	0.000	0.0
49	11.8	0.035	1.000	0.018	0.2
Mean 35–42		0.799	0.841	0.743	8.2
Mean 43–49		0.012	0.151	0.009	1.7

only 8 sampled training snapshots rather than all 34. The training curve plateaus at epoch 30, consistent with the GRU failing to learn smooth weight dynamics from sparse snapshots (Fig. 8).

VII. PRIOR WORK COMPARISON

Table IV compares our results to published Elliptic work. All prior studies use GPU hardware and transductive evaluation. Our MLP (F1=0.744) matches Alarab et al. (F1=0.740) under stricter evaluation; our GraphSAGE (F1=0.697) falls 5.3pp below Lo et al. (F1=0.750), largely due to the transductive-vs-inductive protocol gap.

VIII. DISCUSSION AND CONCLUSION

Three practical implications. First, periodic retraining is essential: a $39\times$ fraud-rate decline cannot be addressed by

TABLE IV
PRIOR ELLIPTIC WORK VS. THIS STUDY

Work	Model	F1	HW	Protocol
Weber [1]	GCN	0.700	GPU	Transductive
Alarab [2]	GCN+feat	0.740	GPU	Transductive
Lo [3]	SAGE	0.750	GPU	Transductive
Pareja [4]	EvolveGCN	0.770	GPU	Temporal
Ours	MLP	0.744$\pm$.005	CPU	Inductive
Ours	SAGE	0.697 $\pm$.009	CPU	Inductive
Ours (ablation)	EvolveGCN	0.290	CPU	Sampled

any static model or post-hoc calibration. Sliding-window retraining is the minimum practical requirement. Second, graph structure is a conditional asset: beneficial in stable periods, harmful under rapid distributional shift. Adaptive monitoring that switches between structural and feature-only prediction could capture both benefits. Third, business cost functions should drive deployment decisions: the highest-F1 model is not always most cost-effective under asymmetric FN/FP penalties.

Limitations. RandomNodeLoader rather than NeighborLoader (missing torch-sparse on M4 CPU) may underestimate GNN performance by 5–10pp. Results are from one dataset; generalisation to other transaction graphs is open. The EvolveGCN result is a hardware lower bound, not a competitive evaluation.

Conclusion. A plain MLP (F1=0.744) outperforms all tested GNNs on the Elliptic Bitcoin Dataset under inductive evaluation. Temporal distribution shift (fraud rate 11.6% $\rightarrow$ 0.3%) causes near-complete recall failure for all models. Feature importance (82.4% local feature dominance) explains MLP competitiveness. Temperature scaling confirms a structural root cause ($\Delta F1=0$). These findings motivate periodic retraining and cost-function-driven model selection for production fraud detection.

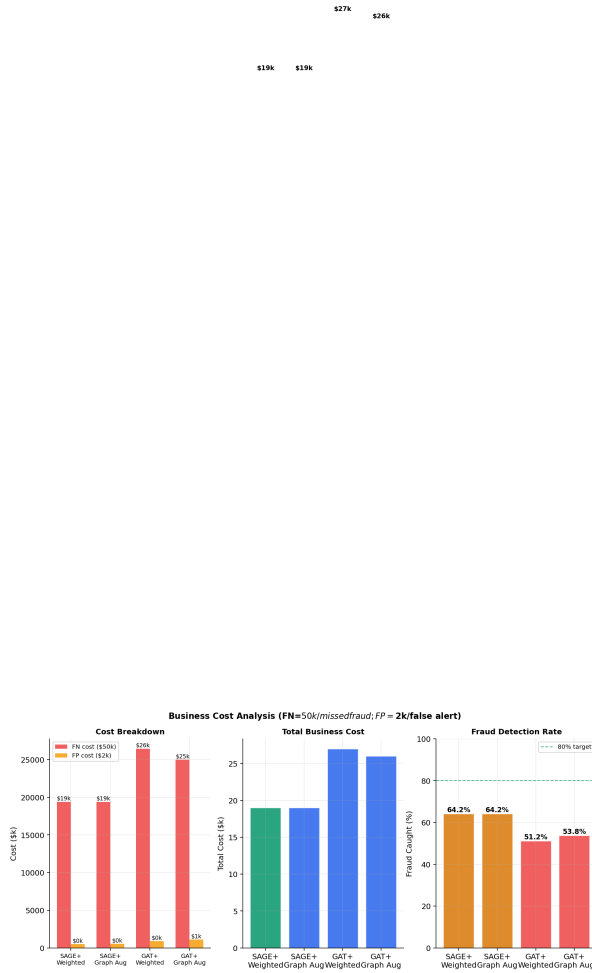


Fig. 6. Business cost analysis (FN=\$50k, FP=\$2k). GAT + Graph Aug. achieves the lowest operational cost despite lower aggregate F1, because catching more fraud outweighs the extra false-alarm cost.

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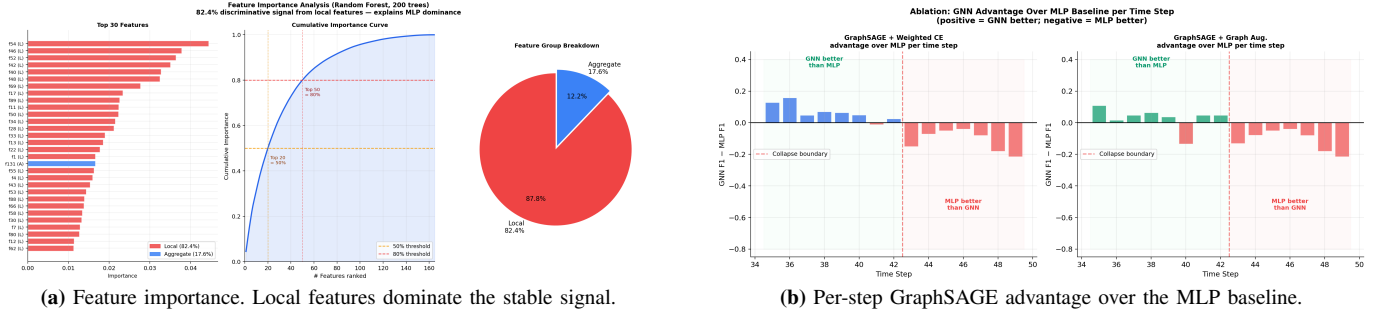


Fig. 7. Analysis pair. Left: Random Forest importance shows that most discriminative power sits in local transaction features. Right: the graph-model advantage reverses after step 42, which is consistent with learned topology becoming stale under temporal shift.

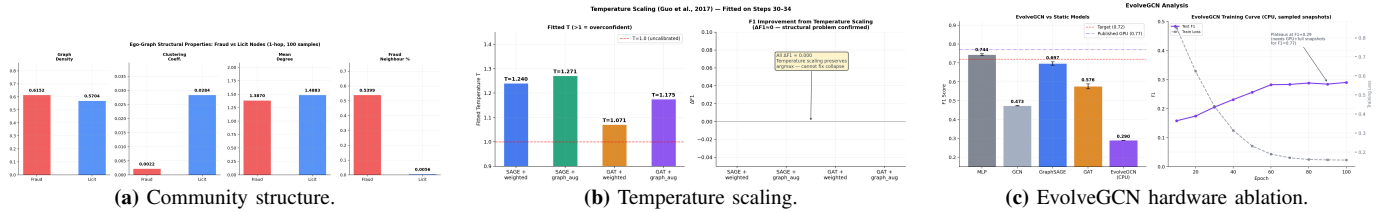


Fig. 8. Three supporting analyses. Fraud communities are real but temporally fragile, temperature scaling does not recover performance, and temporal GNNs remain highly sensitive to hardware and snapshot availability.